INCH-POUND

A-A-54650
15 May 1992

COMMERCIAL ITEM DESCRIPTION

CRANIOTOME SET, Neurosurgical Air Powered

The General Services Administrations has authorized the use of this commercial item description.

This commercial item description covers an air powered neurosurgical craniotome set used for excision of a part of the skull.

Salient Characteristics:

General. The craniotome set shall be air powered, consisting of at least the following components:

1 Air Driven Motor, 20,000 RPM.
1 800/1000 RPM Neuro Modular Attachment (a speed reducing torque converter)
1 Swiveling Dura Guard
1 Skull Perforator Guard
1 Skull Perforator Guard Adaptor (for 1000 RPM drive) (if necessary)
1 Wire Pass Drill Guard
1 Sterilization Case
1 Air Hose, 10 ft. (minimum)
1 Stabilizing and 1 Pistol Grip Handle (Optional)
1 Instrument lubricant
5 Dura Guard O-Rings
5 Neuro Attachment O-Rings

Air Driven Motor.

The driver handpiece (motor), used specifically for neurosurgical procedures shall have the following characteristics:

The motor shall weigh a maximum of 20 ounces and shall have a speed of 0-20,000 RPM.

AMSC N/A FSC 6515

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The motor shall be approximately 6-7 inches (15-17 cm) in length and shall operate at full RPM speed via connection to nitrogen or dry, filtered hospital grade compressed air at a normal operating pressure of 90-100 psi.

The proximal end shall have a connector which will adapt and securely lock onto the air hose.

The distal end shall securely accommodate the locking collar of the 800/1000 RPM neuro attachment as well as the dura and wire pass drill guard.

The distal end shall be cannulated to accommodate the craniotome blade and the wire pass drill. The distal end shall have a simple spring lock type mechanism to securely seat and firmly lock the blade or drill bit into the air driver.

Variance of the driver speed shall be through a simple but highly responsive trigger mechanism.

The braking time from full speed to stop shall not be more than one second.

Shall have a safety mechanism to prevent accidental engagement.

Neura modular attachment.

The attachment shall weigh not more that 16 ounces and shall be approximately 4-6 inches long (10-15 cm).

The proximal end shall have a knurled collar which will lock onto the distal end of the air driver, the power source for the attachment.

The attachment shall have a Hudson type distal end which will securely seat an automatic skull perforator and other optional Hudson type attachments. A simple spring lock mechanism shall firmly lock the skull perforator and optional Hudson type attachments onto the modular attachment.

The distal end shall be formed to snugly accommodate the component skull perforator guard.

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Swiveling dura guard.

The guard collar shall be knurled and lock securely onto the air driver.

The stem shall swivel 360 degrees following the neuro blade when cutting the skull. This shall eliminate the need to change the hand grip when turning a skull flap.

The distal end shall have a thin integral metal projection with an inward notched tip to correctly seat the craniotome blade.

Wire pass drill guard.

The guard shall protect the surgeon and the surrounding tissue from the rotating spindle of the driver and shall press securely onto the distal end of the air driver.

Skull perforator guard.

The guard shall protect the surgeon and the surrounding tissue from the rotating perforator drill and shall securely press onto the neuro modular attachment.

Skull perforator guard adapter.

Shall be supplied with a skull perforator guard adaptor if the neuro attachment is an 800 RPM unit.

Sterilization case.

The case shall be stainless steel and perforated with hundreds of minute holes on top, bottom and sides for maximum steam penetration of case contents.

The case shall be large enough to accommodate all components listed except the lubricant.

The case shall be divided to separate the air hose from the metal instrument. This division shall enable the hose to be loosely coiled.

The top and bottom surfaces shall be flat to faciliate stacking during storage.

The case shall have a removable top with at least two simple flat style closure mechanisms for securing lid during sterilization.

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Air hose.

The air hose shall be 10 ft. (3.05m) quick disconnect coaxial rubber air hose. The distal end of the hose shall be compatible with the proximal end of the air driver. The proximal end of the hose shall have a standard Schrader male connector for insertion into standard Compressed Gas Association (CGA) nitrogen regulator fitting. Air shall exhaust through an air diffuser at the proximal end of the hose (10 ft. from the surgical site).

Lubricant.

The bottle of lubricant supplied shall be specifically for air powered instruments.

O-Rings.

Shall be furnished with at least five (5) dura guard replacement O-rings and five (5) neuro attachment replacement O-rings.

The rings shall securely fit into the locking collar of the instrument, shall be free of nicks and imperfections and shall be color coded for the specific instrument it fits.

All metal parts shall be fabricated of a suitable corrosion resistant metal alloy that has been properly hardened and passivated.

All items in the set, except for the lubricant, shall be capable of sterilization by ethylene oxide or steam up to 275°F.

Operating instructions.

The craniotome set shall be furnished with at least two copies of instructions for operation, maintenance and repair.

Corrosion resistance. The sterilization case and sterilizable items shall show no evidence of corrosion when tested in accordance with ASTM Designation F1089.

Workmanship. The craniotome set shall be free from defects which detract from its appearance or impair its serviceability. Set components shall be free from pits, flash, scale, deformity and foreign matter.

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Unit. Set (SE). One craniotome, as specified, constitutes one unit.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as specified herein. Unit of product for examination purposes shall be one craniotome set. Sampling shall be inspection level II with an Acceptable Quality Level (AQL) of 1.0.

Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified herein. Unit of product for test purposes shall be one craniotome set. Sampling shall be inspection level S-2 with an AQL of 1.0.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

NOTE 1: The components in the craniotome set have the following National Stock Numbers:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6515-01-223-1518	Swiveling Dura Guard
6515-01-139-7554	Wire Pass Drill Guard
6515-01-267-6115	Air Hose, 10 ft.
6515-01-168-2599	Craniotome blade
6515-01-267-6114	Nitrogen Regulator Fitting

NOTE 2: The following National Stock Number and its respective Item Identification is covered by this document:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6515-01-267-1448	CRANIOTOME SET, Neurosurgical, Air Powered

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project: 6515-5308
Location: Cranioto/S-11

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-54650	15 May 1992

3. DOCUMENT TITLE
CRANIOTOME SET, Neurosurgical Air Powered

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Personnel Support Center ATTN: DPSC-MSE	(1) Commercial (215) 737-2870	(2) AUTOVON 444-2870
c. ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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